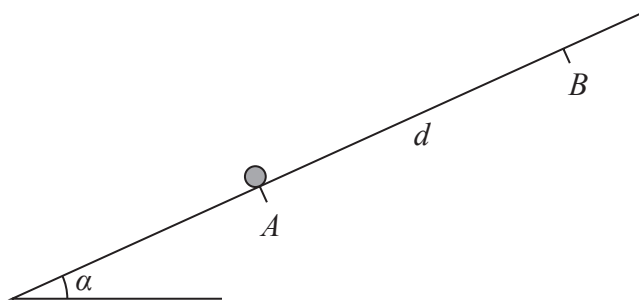


1.



**Figure 1**

A particle of mass  $m$  is held at rest at a point  $A$  on a rough plane.

The plane is inclined at an angle  $\alpha$  to the horizontal, where  $\tan \alpha = \frac{5}{12}$

The coefficient of friction between the particle and the plane is  $\frac{1}{5}$

The points  $A$  and  $B$  lie on a line of greatest slope of the plane, with  $B$  above  $A$ , and  $AB = d$ , as shown in Figure 1.

The particle is pushed up the line of greatest slope from  $A$  to  $B$ .

- (a) Show that the work done against friction as the particle moves from  $A$  to  $B$  is  $\frac{12}{65}mgd$  (3)

The particle is then held at rest at  $B$  and released.

- (b) Use the work-energy principle to find, in terms of  $g$  and  $d$ , the speed of the particle at the instant it reaches  $A$ . (4)



2. A vehicle of mass 450 kg is moving on a straight road that is inclined at angle  $\theta$  to the horizontal, where  $\sin \theta = \frac{1}{15}$

At the instant when the vehicle is moving **down** the road at  $12 \text{ m s}^{-1}$

- the engine of the vehicle is working at a rate of  $P$  watts
- the **acceleration** of the vehicle is  $0.5 \text{ m s}^{-2}$
- the resistance to the motion of the vehicle is modelled as a constant force of magnitude  $R$  newtons

At the instant when the vehicle is moving **up** the road at  $12 \text{ m s}^{-1}$

- the engine of the vehicle is working at a rate of  $2P$  watts
- the **deceleration** of the vehicle is  $0.5 \text{ m s}^{-2}$
- the resistance to the motion of the vehicle from non-gravitational forces is modelled as a constant force of magnitude  $R$  newtons

Find the value of  $P$ .



3. A particle  $P$  moves on the  $x$ -axis.

At time  $t = 0$ ,  $P$  is instantaneously at rest at  $O$ .

At time  $t$  seconds,  $t > 0$ , the  $x$  coordinate of  $P$  is given by

$$x = 2t^{\frac{7}{2}} - 14t^{\frac{5}{2}} + \frac{56}{3}t^{\frac{3}{2}}$$

Find

(a) the non-zero values of  $t$  for which  $P$  is at instantaneous rest (3)

(b) the total distance travelled by  $P$  in the interval  $0 \leq t \leq 4$  (3)

(c) the acceleration of  $P$  when  $t = 4$  (3)

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4. A particle  $P$  of mass  $0.75\text{ kg}$  is moving with velocity  $4\mathbf{i}\text{ m s}^{-1}$  when it receives an impulse  $\mathbf{J}\text{ N s}$ . Immediately after  $P$  receives the impulse, the speed of  $P$  is  $8\text{ m s}^{-1}$

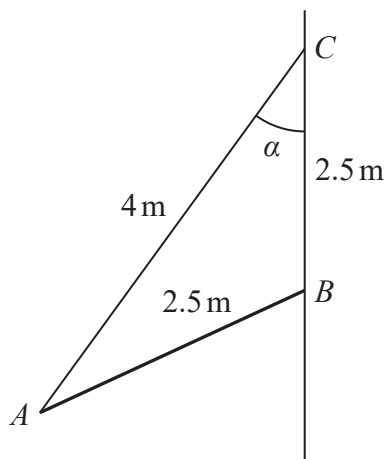
Given that  $\mathbf{J} = c(-\mathbf{i} + 2\mathbf{j})$ , where  $c$  is a constant, find the two possible values of  $c$ .

(6)

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5.



**Figure 2**

A pole  $AB$  has length  $2.5\text{ m}$  and weight  $70\text{ N}$ .

The pole rests with end  $B$  against a rough vertical wall. One end of a cable of length  $4\text{ m}$  is attached to the pole at  $A$ . The other end of the cable is attached to the wall at the point  $C$ . The point  $C$  is vertically above  $B$  and  $BC = 2.5\text{ m}$ .

The angle between the cable and the wall is  $\alpha$ , as shown in Figure 2.

The pole is in a vertical plane perpendicular to the wall.

The cable is modelled as a light inextensible string and the pole is modelled as a uniform rod.

Given that  $\tan \alpha = \frac{3}{4}$

(a) show that the tension in the cable is  $56\text{ N}$ .

**(4)**

Given also that the pole is in limiting equilibrium,

(b) find the coefficient of friction between the pole and the wall.

**(6)**



6. Two particles,  $A$  and  $B$ , are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.  
The mass of  $A$  is  $2m$  and the mass of  $B$  is  $3m$ .

Immediately **after** the collision,  $A$  and  $B$  are moving in opposite directions with the same speed  $v$ .

In the collision,  $A$  receives an impulse of magnitude  $5mv$ .

- (a) Find the coefficient of restitution between  $A$  and  $B$ .

(6)

After the collision with  $A$ , particle  $B$  strikes a smooth fixed vertical wall and rebounds.  
The wall is perpendicular to the direction of motion of the particles.

The coefficient of restitution between  $B$  and the wall is  $f$ .

As a result of its collision with  $A$  and with the wall, the total kinetic energy lost by  $B$  is  $E$ .  
As a result of its collision with  $B$ , the kinetic energy lost by  $A$  is  $2E$ .

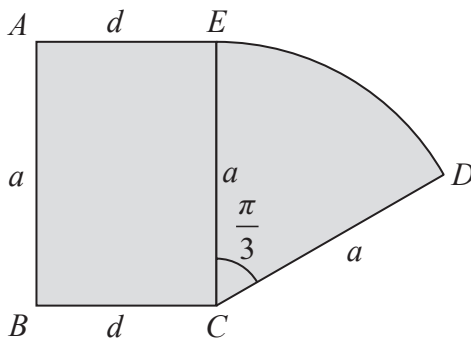
- (b) Find the value of  $f$ .

(4)

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7. In this question you may use, without proof, the formula for the centre of mass of a uniform sector of a circle, as given in the formulae book.



**Figure 3**

The uniform lamina  $ABCDE$ , shown shaded in Figure 3, is formed by joining a rectangle to a sector of a circle.

- The rectangle  $ABCE$  has  $AB = EC = a$  and  $AE = BC = d$
- The sector  $CDE$  has centre  $C$  and radius  $a$
- Angle  $ECD = \frac{\pi}{3}$  radians

The centre of mass of the lamina lies on  $EC$ .

- (a) Show that  $a = \sqrt{3}d$  (4)

The lamina is freely suspended from  $B$  and hangs in equilibrium with  $BC$  at an angle  $\beta$  radians to the downward vertical.

- (b) Find the value of  $\beta$  (7)



8.

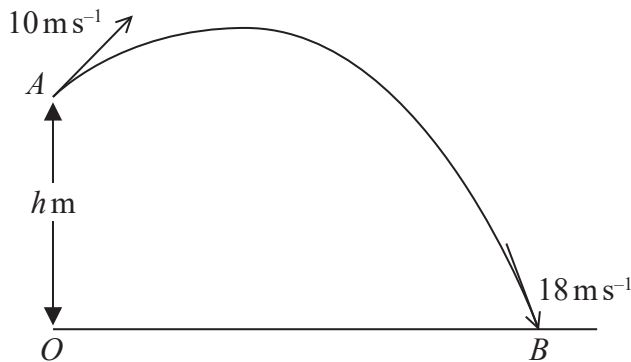


Figure 4

The fixed point  $A$  is  $h$  metres vertically above the point  $O$  that is on horizontal ground. At time  $t = 0$ , a particle  $P$  is projected from  $A$  with speed  $10 \text{ m s}^{-1}$ . The particle moves freely under gravity. At time  $t = 2.5$  seconds,  $P$  strikes the ground at the point  $B$ . At the instant when  $P$  strikes the ground, the speed of  $P$  is  $18 \text{ m s}^{-1}$ , as shown in Figure 4.

(a) By considering energy, find the value of  $h$ . (3)

(b) Find the distance  $OB$ . (5)

As  $P$  moves from  $A$  to  $B$ , the speed of  $P$  is less than or equal to  $8 \text{ m s}^{-1}$  for  $T$  seconds.

(c) Find the value of  $T$  (6)